

ECE 1000 Explorations in ECE

Lab 2: Capacitors & Inductors

September 17, 2025

AIEB 368

Dr. Van Neste ~ cvanneste@tntech.edu

JC Williams ~ jcwilliams@tntech.edu

Project Objectives:

The following are the objectives for the second ECE 1000 lab:

- Follow Part 1 of this handout to build a parallel-plate capacitor.
- Follow Part 2 of this handout to build an inductor.
- Follow Part 3 of this handout to enter the measured and estimated values of capacitance of inductance for the components structured, along with the percent error. Submit the last page of this document to the ILearn assignment (Lab 2).

Background Information:

Through this project, students will become familiar with two common passive components: capacitors and inductors. Both store energy—capacitors in an electric field and inductors in a magnetic field. Capacitance (measured in Farads) describes how much charge a capacitor holds per volt, and inductance (measured in Henries) describes how much magnetic flux is produced per ampere. In practice, capacitors oppose sudden changes in voltage, while inductors oppose sudden changes in current. Although these components are readily available from multiple suppliers (e.g., Digikey, Mouser, Jameco, Amazon), students will build these components with everyday items to develop a deeper understanding!

Part 1: Building a Capacitor:

- Each student has been given one sheet of copy paper, two pieces of aluminum foil, and tape.
- Fold the copy paper in half (“hamburger style”) so the dimensions are $5.5'' \times 4.25''$.
- Tear the copy paper along the fold.
- These half-sheets will serve as the dielectric for the capacitor (see Figure 1).

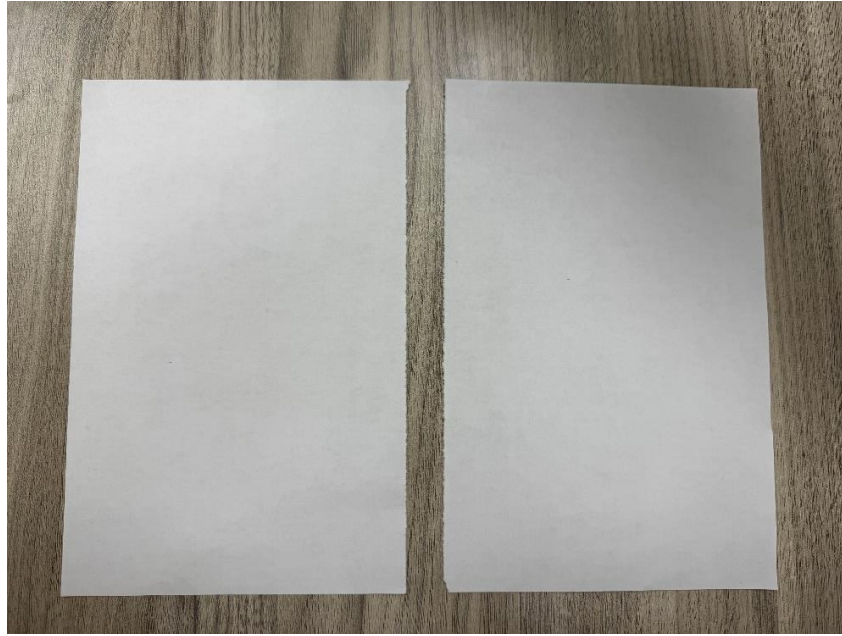


Figure 1: Split Copy Paper Used as the Capacitor Dielectric

- Cut the aluminum foil in half to make two rectangles, each roughly the same size as a half-sheet of the copy paper (about $5.5'' \times 4.25''$).
- Center an aluminum-foil rectangle on each half-sheet and tape it in place (see Figure 2).



Figure 2: Aluminum Foil Plates on Copy-Paper Dielectric

- From the leftover foil, cut two small squares (about $1'' \times 1''$).
- Tape one square to the upper-left corner of one sheet and the other to the upper-right corner of the other sheet (see Figure 3).



Fig. 3. Capacitor Terminals Placed on Opposite Corners

- Stack the two sheets, foil sides facing inward, and tape them together to form a parallel-plate capacitor with the copy-paper dielectric between the plates (see Figure 4).
- Using a paper thickness (d) of 0.10 mm, a relative permittivity (ϵ_r) of 2.5, and the provided plate area value (A), calculate the expected capacitance (C) of the capacitor (use consistent units)!

$$C = \frac{\epsilon_r \epsilon_0 A}{d}$$

- Measure the capacitance (C).
- Roll the capacitor from one end to increase the capacitance (C); ensure the foils do not touch!
- Measure the capacitance (C) again and compare it to the flat configuration (see Figure 5).



Fig. 4. Completed DIY Parallel-Plate Capacitor Made from Copy Paper and Aluminum Foil



Fig. 5. Rolled DIY Parallel-Plate Capacitor

Part 2: Building an Inductor:

- Each student has been given a nail and insulated copper (magnet) wire. Wrap the insulated wire neatly around the nail from one end to the other end in a single layer (see Figure 6).

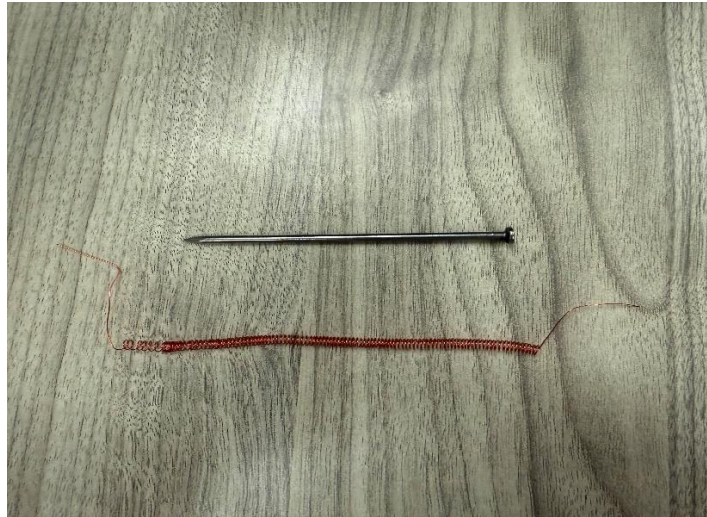


Figure 6: Magnet Wire Wrapped Around Nail

- Record the total number of turns wound on the nail (from bottom to top).
- Using sandpaper, strip 1" of insulation from each end until copper is exposed (see Figure 7).

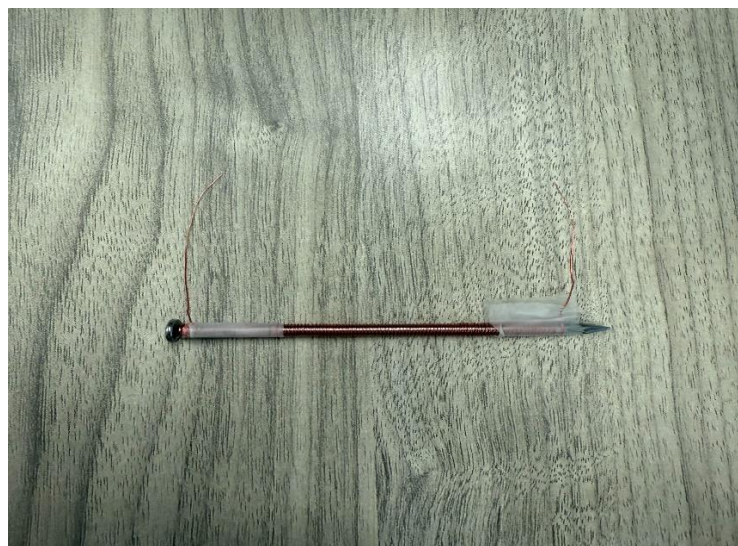


Figure 7: Inductor with Nail Core and Insulated Magnet Wire

- With the power supply **OFF**, connect the coil to the supply. Set the voltage to 5 Volts and the current limit to 0.25 Amps, then enable the output. **CAUTION: DON'T TOUCH THE BARE COPPER WIRE WHILE ENERGIZED (only touch insulated material).**
- Verify operation by hovering the nail over small metallic objects to test the electromagnet.
- Using the turns (N), a nail relative permeability (μ) of 100, a nail length (l) of **3.5"**, and a nail radius (r) of 2 mm, calculate the expected inductance (L) of the inductor (careful of units)!

$$L = \frac{\mu_r \mu_0 N^2 A}{l} = \frac{\mu_r \mu_0 N^2 \pi r^2}{l}$$

Part 3: Measurements:

Capacitance:

Signature: _____

Capacitance (C)	Enter Measured Capacitance	Enter Your Expected Capacitance	<i>Percent Error</i> $= \left \frac{M - E}{E} \right * 100$
C1			
C2			
C3			
C4			

Series & Parallel Capacitance:

Signature: _____

Capacitance (C)	Enter Measured Capacitance	Enter Your Expected Capacitance	<i>Percent Error</i> $= \left \frac{M - E}{E} \right * 100$
All Capacitors Connected in Series			
All Capacitors Connected in Parallel			

IF TIME PERMITS, COMPLETE THIS SECTION. IF THERE IS NOT TIME, COMPLETE LATER.

Inductance:

Signature: _____

Inductor (L)	Enter the Capacitance Value of DIY Capacitor	Enter the Resonant Frequency	Enter the Measured Inductance
L1			
L2			
L3			
L4			